



The Ephemeris

June 2016

Volume 27 Number 02 - The Official Publication of the San Jose Astronomical Association

June - Aug 2016 Events

Board & General Meetings

Saturday 6/18, 7/16, 8/20

Board Meetings: 6 - 7:30pm

General Meetings: 7:30-9:30pm

Fix-It Day (2-4pm)

Sunday 6/5, 7/10, 8/07

Solar Observing

Sunday 6/5, 7/10, 8/07

Intro to the Night Sky Class & Talk

Houge Park 1Q In-Town Star Party

Friday 6/10, 7/8, 8/12

Astronomy 101 Class & Talk

Houge Park 3Q In-Town Star Party

Friday 6/24, 7/29, 8/26

RCDO Starry Nights Star Party

Saturday 6/25, 7/23, 8/27

Binocular Star Gazing

Saturday 5/28, 7/23, 8/13

Imaging SIG Mtg

Tuesday 6/21, 7/19, 8/16

Astro Imaging Workshops

(at Houge Park)

Saturday 7/23, 8/27

Quick START (by appointment)

Friday 6/11, Saturday 8/13

*Please refer to the SJAA Meetup Group
to obtain the most up-to-date information
on SJAA events at:*

www.meetup.com/sj-astronomy

SJAA events are subject to cancellation due to weather.

Please visit website for up-to-date info.

Cover Photo: IC 1805

SHO Heart

By SJAA Member Rob Pfile

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Guest Speaker Program Update - 2016

From Sukhada Palav

Here's the current line-up of guest speakers:

- ◇ April 23: Topic: Mars, presented by Dr. Chris McKay
- ◇ May 21: Topic: Planetary Nebulae, presented by Dr. David Clark
- ◇ June 18: Topic: to be announced, presented by Francesca Fornasini
- ◇ July 16: Movie Night
- ◇ Aug 20: [open]
- ◇ September 23: Topic: Messier Marathon, presented by Don Machhotz

If you have suggestions for possible speakers, please send name, email address and other info to: Speaker.sjaa@gmail.com

Spring 2016 SJAA Swap Meet

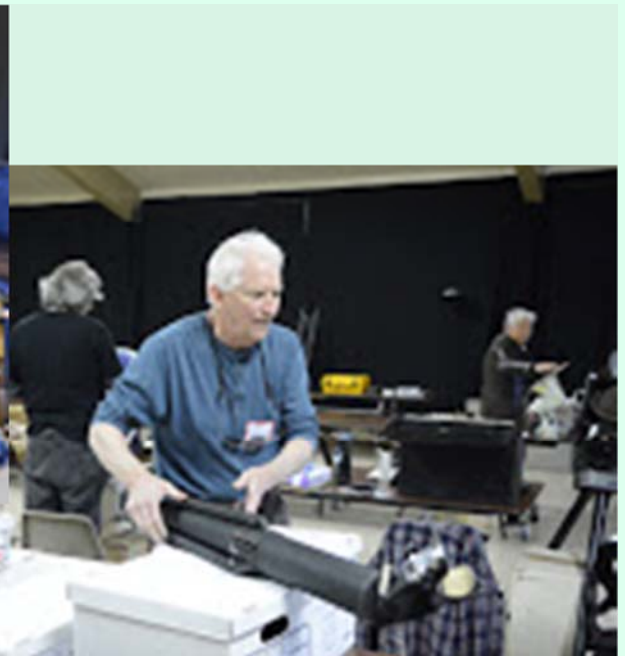
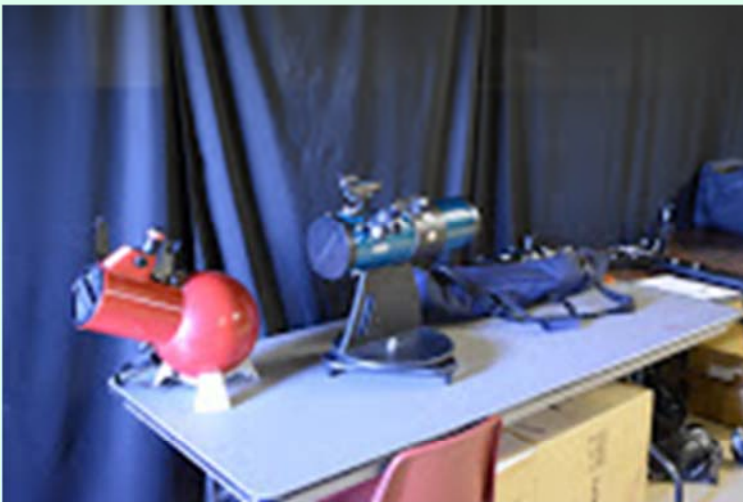
From Rob Jaworski

The SJAA Spring Swap Meet for 2016 was another resounding success! The club was able to release a number of scopes back into the amateur astronomy community, gear from both donations as well as equipment that outlived its useful life in the loaner program.

At the swap meet last fall, the volunteers and board members realized that the swap meet is just as effective in getting used equipment to change hands as the annual auction. The main difference is that the swap meet is far less energy and labor intensive, for both the volunteers and attendees. A poll was put out after the fall swap meet to ask the community if there was a strong preference for an auction in addition to or over a swap. All responses indicated that holding a swap meet, and foregoing an auction, would be entirely acceptable. So that's what happened in March: Instead of the annual auction, a swap was held, and it was just as successful.

The original idea was to hold three swap meets a year, with potentially the next happening sometime in July. However, at April's board meeting, a straw poll was held among the board and others present, and the collective thinking was that two a year is plenty. So we'll do this again in the fall.

The board appreciates everybody that responded to the poll, as well as those members who participated at the swap meet. We hope everyone had a good time, was able to meet their objectives of finding new, or divesting old, gear, and still get the opportunity to see and meet other astronomers in full daylight!



*Swap Meet Photo Credits:
Ed Wong*

Transit of Mercury - May 9th, 2016

Thanks to event sponser Teruo Utsumi Houge Park was ready at 6:30 AM with plenty of spirit, hot chocolate and donuts to go around. The clouds almost got in the way. But not quite. The next Mercury transit will occur on November 11, 2019.

Photo: Gary Chock

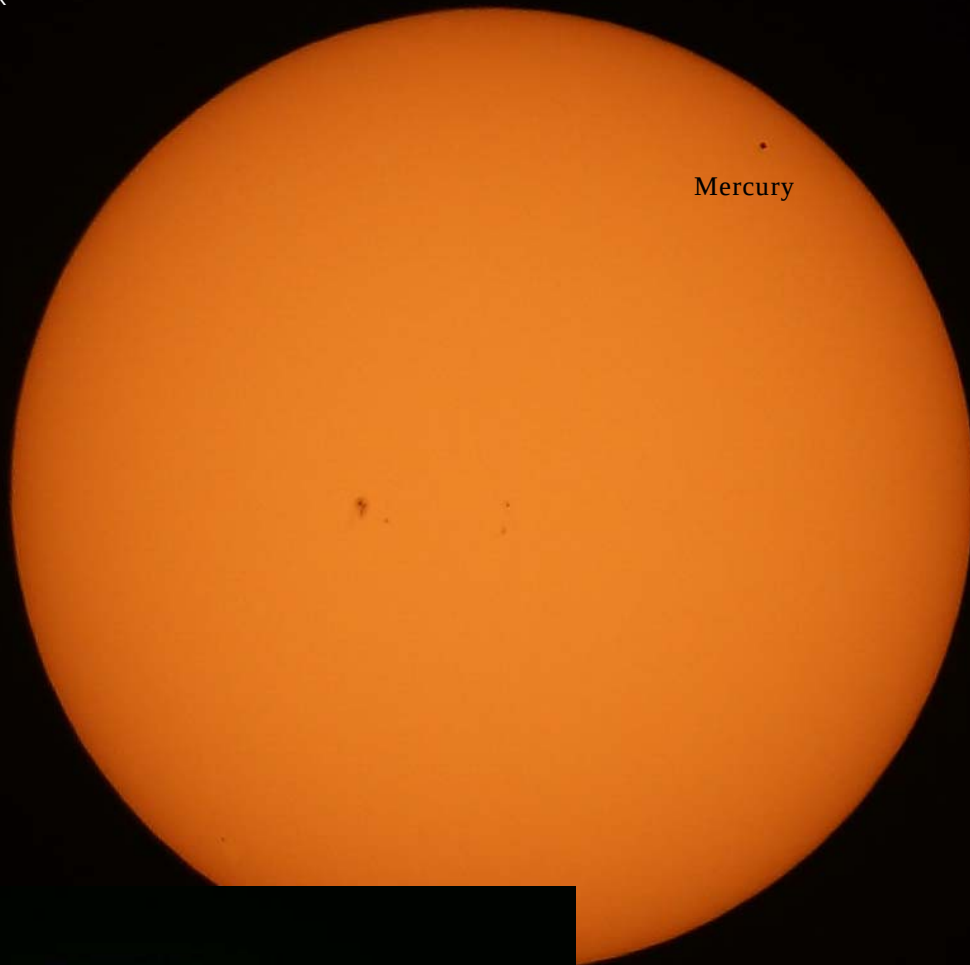


Photo credit: Julia



Photo credit: Julia



Grand Unified SJAA, Feb 2016

From Teruo Utsumi

I've dubbed our automation effort GUS: Grand Unified SJAA. (This is a continuation from the post in May. If you're interested in pitching in or just curious, you're invited to join the Coders group.)

The organizers spend hundreds of man hours every year in order to streamline our various processes, we've started the task of automating our major operations:

Membership — We've grown quite a bit in recent years. The SJAA membership is now about 350, so you can imagine the gymnastics needed to juggle that many on a spreadsheet. Our primary options are 1) Simple Machines Forum (SMF) + Paypal and 2) Tendenci. The board of directors has already been using SMF to conduct club business so it's a natural fit. We just started looking at an open source membership management package called Tendenci. The plan is to eventually allow all members access to an SJAA specific forum, all under **www.sjaa.net**.

Events Believe it or not, we hold roughly 150 distinct events every year. Since we mainly run off the lunar cycle and schedule around twilight, it's not as simple as scheduling a recurring event on the same week of every month at the same time. We're designing a system that will automatically:

- generate all recurring events for a calendar year,
- update our Google calendars,
- post events on Meetup,
- send email announcements, and
- (perhaps sometime in the not too distant future) post to Twitter and Facebook.

We're also looking to automate management of the various tasks associated with each of our events.

Library/equipment loaner programs Most of the work here is bookkeeping, tracking the various assets. Automation will streamline the process and allows online access to the information.

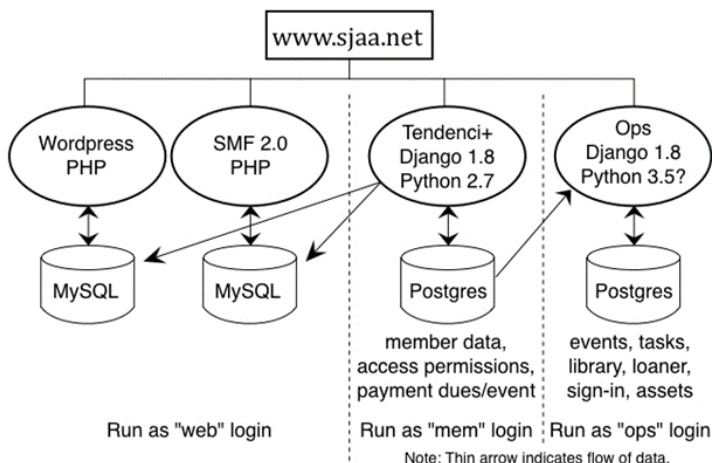
Sign-in In the past year we started using paper sign-in sheets for our events. This allows us to provide documentation to the city of San Jose which grants us free access to the facilities we enjoy at Houge Park. A digital sign-in system will allow us to take a detailed look at who's coming to our events and help us to better understand how to reach out to our members as well as the general public.

The Coders Group is currently made up of about a dozen people. We'd like to extend our appreciation for all Coders for their participation and help.

Special thanks to:

Swami Nigam - set up of SMF
Thomas Stucky - library/loaner app coding
Murali Balasubramaniam - sign-in app coding
John Wainwright - hosting Tendenci during our evaluation, expertise in all things SW
Guillermo Jimenéz - looking into Tendenci
Satish Vellanki - work on club website
Teruo Utsumi - event coding, GUS architecture

We'll have updates as we hit future milestones. We're just getting started, but the system might end up looking something like the flow chart below:



Lens Cleaning

From Mark Striebeck

After almost 3 years with my TOA-130 scope, there was quite a layer of goop on the front lens (also, my dew heaters didn't work - recently switched to DewBuster - hopefully that will work). So, I decided to do some cleaning.



I used instructions from Astro-Physics, see details on my blog: listed below. Now, I have to wait for good seeing to find out if I was super careful enough.

<http://mstriebeck-astrophotography.blogspot.com/2016/04/cleaning-my-front-lens.html>

Photo Credits: Mark Striebeck

GPS: Global Positioning on a Shoestring March 2006 Ephemeris

From Bill Maney

How do you figure out where you are on the planet? These days you can find your address on a mapping website, or maybe switch on a GPS. But before these things what could you do?

When I got into astronomy I also started looking at celestial navigation techniques. I got a cheap student sextant (\$35) and a nautical almanac and taught myself how to take a sight and convert it to a line-of-position on a map. With a few lines plotted, I could pinpoint myself within a handful of miles. But it was complicated and I was left asking: If I know what star is right above me, isn't that enough? That same star isn't above any other place on the planet, so that's the only measurement I should need.

So I made an experiment. This is where the shoestring comes in. I lay down on something soft on my driveway and hung a shoestring from a basketball hoop. OK so it was a regular piece of string. I got it wet so that it would hang straighter. I mentally extended the line of the string into the sky and noted the point that it hit and also noted the time. Drops of water got me directly in the eye, so I knew I was well aligned.

Then I dragged out my star hopping skills and found the spot on my star chart. The declination of that point is more-or-less the latitude of my driveway. (If I did my math correctly, the non-roundness of the earth causes the measurement to be 5 minutes (= 5 nautical miles) lower than my actual latitude).

The right ascension is not my longitude of course because the earth spins. The longitude is given by a formula that I worked out by trial and error using data from www.nao.rl.ac.uk and is not super accurate (within a half mile or so for 2006) and will probably be pushed into the online version of the newsletter by my editors. It goes as follows, where Tx is the number of hours since midnight, Jan 1, 2006, Greenwich mean time (subtract 8 hours for the number of hours since midnight PST or 7 hours for PDT), and 6.700555556 was the Right Ascension of the sky over Greenwich at that moment, and Rax is the Right Ascension measurement from the driveway, and Lyear = 365.256363, the length of a year, and frac returns the fractional part of a number. $\text{Longitude} = 15 * (\text{Rax} - 6.700555556 - 24 * \text{frac}(\text{Tx} * (\text{Lyear} + 1) / \text{Lyear}) / 24)$ Negative numbers are W latitude. If you get a number larger than 180, subtract or add 360.

I had to do the experiment a few times to get it right. On the

last try, I put in the time to the nearest minute and was only 5 miles off in longitude. Then I corrected it with the time to the nearest second and I was only 1 mile off! This is unbelievable accuracy...literally. That this was pure luck is evidenced by my 3 tries for latitude which were between 15 and 55 miles off. Hey, it's a big planet! And what did you expect for a shoestring!?!

How useful is this? Well... Good luck using it on a rocking boat. And on land, if you don't know where you are within 55 miles, you're pretty lost. The accuracy could be improved by using some magnification instrument. A design using a spinning vat of mercury (highly toxic) is quite appealing (and far from being executed). I'll leave it as an exercise to the reader to determine the point of all this. If you try this technique yourself, the usual warnings about laying down on driveways in the dark apply.

Editor's Note: I found this article from Bill Maney ten years back in the Ephemeris archives and thought it was pretty interesting. I am not sure how Mr. Maney developed his formula but it seems very creative.

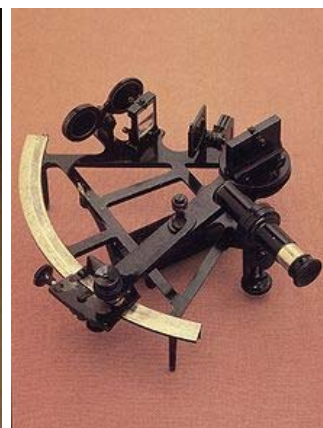
Determining the longitude of a ship sailing at sea was historically the holy grail of mariners and reasonable method of calculation was not achieved until the mid-seventeen hundreds. The British Board of Longitude and subsequently the British Royal Navy adopted two different methods; the first was the Lunar Distance method, which involved Lunar sightings and very labor intensive calculations and then later as clock technology improved, the Marine Chronometer, which allowed use of an accurate but portable time standard in conjunction with celestial navigation to determine the longitude. Piller

Reference Source: Wikipedia.



Harrison's "Sea Watch" No.1 (H4), with winding crank

Source: Wikipedia



Sextant pictured above.

The primary use of a sextant is to determine the angle between an astronomical object and the horizon for the purposes of celestial navigation.

Source: Wikipedia

Cupertino Earth Day & Solar Party

April 30, 2016



Saturday, April 30, was Earth Day at Cupertino's Civic Plaza. Wolf, Emil, Gary, Bill, plus Marion & Paul all brought solar telescopes and we had a huge number of visitors.



Spacecraft Juno set to arrive at Jupiter on July 4

Juno was launched on Aug. 5, 2011. The spacecraft will orbit the Jovian world 33 times, skimming to within 3,100 miles (5,000 kilometers) above the planet's cloud tops every 14 days. During the flybys, Juno will probe beneath the obscuring cloud cover of Jupiter and study its aurorae to learn more about the planet's origins, structure, atmosphere and magnetosphere.

Juno will make two 53-day elliptical laps around Jupiter, before beginning the mission's science phase. At that point, the spacecraft will begin orbiting the Jovian world every 14 days, from a distance as close as 3,100 miles (5,000 kilometers). It will peer beneath Jupiter's cloud tops to learn about the planet's origins, composition and magnetosphere. Jupiter lies in the harshest radiation environment in our solar system, so this particular spacecraft orbit insertion will mark a new achievement in planetary exploration.

Juno's name comes from Greek and Roman mythology. The god Jupiter drew a veil of clouds around himself to hide his mischief, and his wife -- the goddess Juno -- was able to peer through the clouds and reveal Jupiter's true nature.



Launching from Earth in 2011, the Juno spacecraft will arrive at Jupiter in 2016 to study the giant planet from an elliptical, polar orbit.

Credits: NASA/JPL-Caltech

Promising Worlds Found Around Nearby Ultra-cool Dwarf Star

Astronomers using the TRAPPIST telescope at ESO's La Silla Observatory have discovered three planets with sizes and temperatures similar to those of Venus and Earth, orbiting an ultra-cool dwarf star just 40 light-years from Earth.

TRAPPIST-1 is an ultra-cool dwarf star — it is much cooler and redder than the Sun and barely larger than Jupiter. Despite being so close to the Earth, this star is too dim and too red to be seen with the naked eye or even visually with a large amateur telescope. It lies in the constellation of Aquarius (The Water Carrier).



This artist's impression shows an imagined view of the three planets orbiting an ultra-cool dwarf star just 40 light-years from Earth that were discovered using the TRAPPIST telescope at ESO's La Silla Observatory. In this view, one of the inner planets is seen in transit across the disc of its tiny and dim parent star.

Credits: ESO/M. Kornmesser/N. Risinger (skysurvey.org)

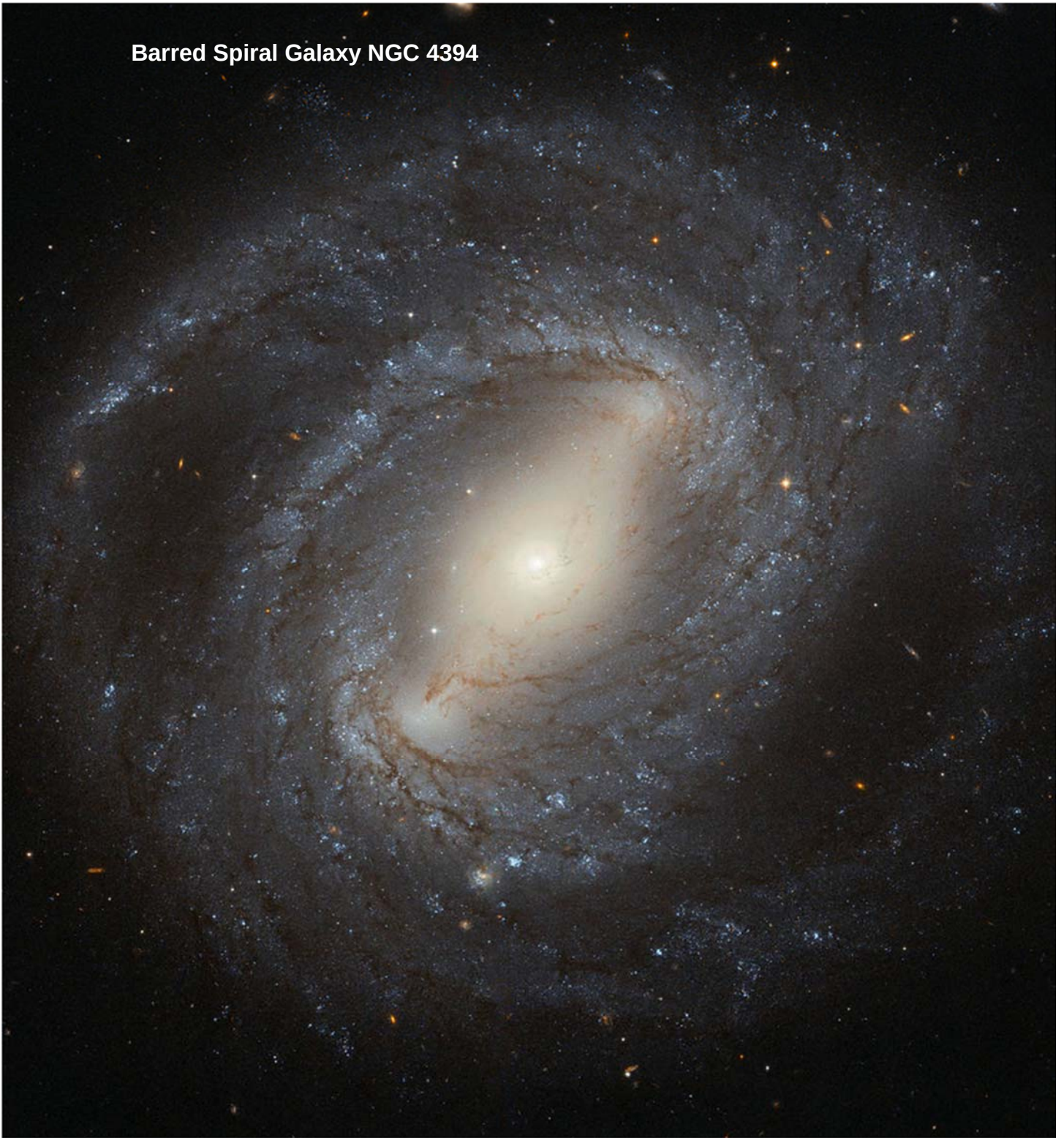
Follow-up observations with larger telescopes, including the HAWK-I instrument on ESO's 8-metre Very Large Telescope in Chile, have shown that the planets orbiting TRAPPIST-1 have sizes very similar to that of Earth. Two of the planets have orbital periods of about 1.5 days and 2.4 days respectively, and the third planet has a less well-determined orbital period in the range 4.5 to 73 days.

"With such short orbital periods, the planets are between 20 and 100 times closer to their star than the Earth to the Sun. The structure of this planetary system is much more similar in scale to the system of Jupiter's moons than to that of the Solar System," explains Michaël Gillon.

Although they orbit very close to their host dwarf star, the inner two planets only receive four times and twice, respectively, the amount of radiation received by the Earth, because their star is much fainter than the Sun. That puts them closer to the star than the so-called habitable zone for this system, defined as having surface temperatures where liquid water can exist, although it is still possible that they possess potentially habitable regions on their surfaces. The third, outer, planet's orbit is not yet well known, but it probably receives less radiation than the Earth does, but maybe still enough to lie within the habitable zone.

Credit: NASA

Barred Spiral Galaxy NGC 4394



Discovered in 1784 by the German–British astronomer William Herschel, NGC 4394 is a barred spiral galaxy situated about 55 million light-years from Earth. The galaxy lies in the constellation of Coma Berenices (Berenice's Hair) and is considered to be a member of the Virgo Cluster.

NGC 4394 is the archetypal barred spiral galaxy, with bright spiral arms emerging from the ends of a bar that cuts through the galaxy's central bulge. These arms are peppered with young blue stars, dark filaments of cosmic dust, and bright, fuzzy regions of active star formation. At the center of NGC 4394 lies a region of ionized gas known as a low-ionization nuclear emission-line region (LINER). LINERs are active regions that display a characteristic set of emission lines in their spectra— mostly from weakly ionized atoms of oxygen, nitrogen and sulphur.

Text credit: European Space Agency

Image credit: ESA/Hubble & NASA, Acknowledgement: Judy Schmidt

SJAA Member Observation Posts

Herschel II from a Red Zone

From Mark McCarthy

Because of the cloudy weather this winter I have not been able to get to darker sites as often as I would wish. I've been working on the Astronomical League's Herschel II and Planetary Nebula lists, but have fallen behind. All the "next up" targets set soon after dark. In order to keep pace I set up my 20-inch telescope in my red zone backyard two clear weekday nights during the April new moon. SQML ranged from 18.3 - 18.5, which equates to Naked Eye Limiting Magnitude of 4.2 to 4.4. The experience was a lesson in how light pollution affects visual observation of deep sky objects.

Surprisingly all but the most diffuse planetary nebula were viewable, as were the open clusters. Of course in a darker sky these would appear brighter, with more detail, color, and stars; but they could be seen nonetheless. The galaxies, however, were hit hard. The galaxies on the H2 list have integrated magnitudes of 13.0 or so, and can be seen well in a 10-inch telescope at dark site. In the 20-inch in a suburban-rural transition "yellow" zone like Fremont Peak, they are bright and obvious, often yielding subtle detail. In my back yard with the 20-inch they were threshold objects for which I needed to work hard to see. If I hope to keep up with the lists I may continue to observe from the back yard, but I should prepare myself for a frustrating experience.

Here are my notes:

NGC 2259, Open Cluster in Monoceros, 3.5' x 3.5', 10.8v: @ 205x: Pretty arrangement of stars. Bright red star to the West. Bright white in middle with a few fainter stars boxing it in. Three in SW make a triangle pointed NE. Faint stars come into view with averted vision, especially a close pair to SW of the center star. Stars mostly E-W. Did not see the nebulosity which is associated with the cluster. Trumpler class II1pn

NGC 2261, Reflection Nebula in Monoceros, 2.0' x 1.0': @ 205x Hubble's Variable Nebula. Wow, interesting! Looks like a comet with a wide, faint but short tail, with a star as the comet nucle-

us. OIII filter loses the object altogether; UHC dims the point star but elongates the tail. This nebula is illuminated by R Monocerotis, and changes to the amount of nebulosity seen are thought to be caused by dense clouds of dust sometimes block R Mon's light. It will be worth coming back to this over the years and try to note its changes. NGC 2264 the Christmas Tree Cluster is nearby, stopped for a look through the 50mm finder.

NGC 2274, Elliptical Galaxy in Gemini, 13.51B, 1.9' x 1.9': @ 205x: Extremely faint, threshold object. But I can see a small bright round core with a just stellar nucleus; halo is a faint round haze. There is a hint with averted vision of a second hazy core just to the north, slightly elongated NNW-SSE, half the size of 2274 [this turns out to be NGC 2275, 1.5' x 1.1' 13.67B]

NGC 2331, OC in Gem, 19' x 19': @ 121x: Coarse, large cluster, about 30 stars with medium range of brightness. A loose circlet of 5 stars lies to the east. "Y" shape overall, like the Hyades. A red star just SW of center. Trumpler IV2m

NGC 2339, barred spiral Galaxy in Gem: 2.7' x 2.0', 12.47B @ 121x. Very faint, AV only but can hold a brighter core with a suggestion of halo 3:2 N-S. Lies in a fine, rich field of stars.

NGC 2415: Irregular Galaxy in Lynx: 0.9' x 0.9', 12.55B @ 121x: Faint oval glow, with a brighter center, in a line of three stars. A nice close double star is midway to the edge of the FOV to the north.

NGC 2493, Lenticular galaxy with bar, in Lynx, 2.0' x 2.0', 13.09B: @ 121x: Seen direct vision but barely. Averted vision brings out barely oval core with some mottling, and suggestion of round halo. Bright red star to the NE.

NGC 2500, spiral galaxy in Lynx, 2.9' x 2.6', 12.23B: @ 121x: Excessively faint round haze, seen averted vision only. Diffuse -- no core noticeable. In a diamond shaped asterism. It is part of a physical system which includes NGC 2537, 2541, and 2552 which are widely separated from each other.

NGC 2541, spiral galaxy in Lynx, 6.3' x 3.2' 12.25B: @ 121x: Extremely faint streak, small. Averted vision shows stellar nucleus, very diffuse halo 2:1 NNE-SSW. It is the largest of the group of galaxies which includes NGC 2500 as above.

Kohoutek K2-2, Planetary Nebula in Mon: @ 205x: I first saw a faint bloated star which I suspected to be the PN. With OIII there was a small faint round shell with a central star -- the shell appeared best while blinking. It is at the base of a mini Big Dipper asterism.

Minkowski M1-7, PN in Gem: @ 205x: Can't believe I got this one in a red zone. Unfiltered appears as a bloated faint star. OIII shows it as a faint, small hazy ball with soft edges, slightly green. No central star seen.

NGC 2371/2372, PN in Gem, 2.2' x 1.0' 13.0B: @ 205x: This is a single physical object, but Herschel first saw it as two faint nebulae, so it received two NGC designations. Unfiltered it appears as an irregular grey glow. With OIII two lobes appear--kissing orbs--both greenish. 2271 is to SW, the fainter 2272 is to NE. Stars appear in both lobes but they are not centrally placed; central star is supposed to be in geometric middle but was not seen.

NGC 2392, "Eskimo" PN in Gem, 0.75' x 0.75' 9.9B @ 205x: Unfiltered can see central star with a round grey glow around it; a close-by star to the N. OIII loses the CS but shows the ragged outer fringe and mottling inside the shell, especially a bright inner shell rim on the NE side. A Herschel discovery.

I made attempts at Abell 21 (aka Medusa Nebula) in Gemini on both nights, but I can't claim it. My sketching showed the correct star field, but I drew a small glow -- when I looked at images after the session I saw its brightest part is a large, thick "C" shape. I suspected a large round but excessively faint glow, but it was so obscure I didn't pin it down as the nebula. Now that I know what to look for I will give it another try.

SJAA Member Observation Posts

New Zealand - March 2016

From Teruo Utsumi

Just got back from a South Island hiking tour in New Zealand.

I took w/ me my 80mm Pentax spotting scope, tripod and 7x42 binocs. The idea was dual day/night time use. For the most part I lugged around the added 15 lbs., but I'd been getting in shape the weeks prior and hiked Mission Peak the day before departure w/ full gear, so I figured I was ready.

There aren't many places where you go from subtropical straight to glaciers. There were plenty of things to do and see. With the binoculars and scope you could see details like glacier crevasses and huge banks of snow just ready to give way. Mt. Cook was particularly spectacular! Through it all, though, having the sun towards the north was a bit odd, something I didn't quite get used to.

From nearby my hotel in Christchurch, it was easy to find the Southern Cross and Alpha/Beta Centauri to the east. Beta and Epsilon Centauri practically point to Omega Centauri. And sure enough toward the west is Orion, but the poor laddie is upside down. That's when I knew I wasn't in Kansas anymore....

The skies were certainly darker than in SJ -- the Milky Way was quite visible, but not quite dark enough to be able to see LMC/SMC. Gemini was also visible, but since they're twins, it's hard to tell if they're upside down.

The first night at a dark sky location I did spot LMC/SMC easily. Another night was spent aboard a ship in Milford Sound for an overnight cruise. Despite the high canyon walls, I was able to see most everything I had seen previously. I "discovered" a globular cluster on the north edge of SMC w/ binoculars. Think I'll name it "NGC 104". I check my app. Wow, the Sky Safari guys are quick -- they already have it in their database. Amazing! The deck light prevents it from being a true dark-sky experience, but I'll take it.

There are stargazing programs (requiring \$\$) in Queenstown (cloudy both nights) and Mt. Cook. I tried to

get to the program at Mt. Cook but didn't get there in time. Instead I found a spot just a few minutes from the motel. It was the best night of the entire trip -- Southern Cross, Centaurus, Orion, Corvus, Gemini, Jupiter, M44, LMC/SMC were naked eye, And shared them w/ a couple tour mates.

Definitely a trip worth taking!

Nice ISS pass

From Rob Jaworski

April 16, 2016 9:09 PM

I love it when I just happen to see an ISS pass that I wasn't expecting. That happened a few minutes ago, on the side of the house, taking out some garbage, when I see a really bright light passing high overhead. No blinking lights, it's gotta be the ISS, I thought.

Heavens Above (HA) confirmed it, at a -2.9 magnitude, passing really close to the star Alioth in the inner handle of the Big Dipper, then toward the SE into Canis Venatici, and passing close to the star Cor Caroli.

What caught my eye, no, what made me think I was seeing things, was that another satellite was heading to the east-north-east, faintly, near Cor Caroli, then fading out into the earth's shadow, just as ISS was in the vicinity. At first, I thought this was a star, but it was moving too, and when I went back to look for it a moment later, it was gone! That whole part of the sky 'under' Cor was entirely absent of any pinpoints of light. That was pretty cool.

HA also indicated that ISS was passing pretty much directly over Virginia City, Nevada, and had an altitude of 51 degrees from my vantage point in San Jose.

I always keep an eye out on the sky, just in case I just happen on an unexpected pass. Tonight's was grand!

Crater Copernicus at 885x

From Mark McCarthy

Following is what I saw the night of March 17, 2016 and it was amazing.

Seeing cycled from good to great in waves. There were 6-8 second waves

where fine detail was fuzzed out, then a 2-3 second calm of very sharp and steady views. There must have been some kind of "wavy" wind pattern coming in regular intervals from the ocean. 885x is 70x per inch of aperture of the 12.5-inch f/7 telescope I used from my back yard-- well beyond the commonly held maximum useful magnification figure of 50x inch. So I guess I can stop worrying about the mirror's quality. It helped too that I was using an equatorial platform, and just needed to look through the eyepiece without disturbing the telescope.

Copernicus was huge, and it took on a weight and body I'd never imagined. Usually it appears flat, and only interesting for its terraces in the crater and the finely granulated particles of its ejecta fanning out across much of the lunar surface. But last night, at 70.84% illuminated, waxing, Copernicus looked like the stump of an ancient tree whose center had rotted out, but whose gnarled, many-branched roots still gripped the lunar surface. The crater walls were *tall*, with sheer cliffs at each terrace level. The rim was bright and curved sinuously. The depth and physical presence of the thing was astonishing. Nearby Eratosthenes too was bulky, as if built up out of clay.

None of my moon books give a photo showing what I saw. *The Kaguya Lunar Atlas* has a very nice image on plate 49, which show the ribbon like crater rim, but it doesn't show the height of the thing. *The Hatfield Photographic Lunar Atlas* plate 5b is pretty good, but the depth is only hinted, and it does not show the ropey hills and ridges which make the outer sides of the crater, which I can only liken to buttress roots around a large tree's trunk. I think Rukl plate 31 comes the closest, though this is a drawing. There's not much sense of the crater's depth in Rukl, but the ropey, veined ridges radiating around the crater are more distinct. It's only a little larger than the field of view I had, too.

A terrific view. Next time the seeing is as steady as that, I'll be sure to pump up the magnification again.

SJAA Member Astrophoto Gallery

Subject: M42 - Orion Nebula

From: Paul Mahany

Location: Yosemite
Date: unknown
OTA: ATM 6" F5 Newtonian
Mount: Atlas EQ 6
Camera: Canon 60Da
Exposure: 4.7 hours
Processing:
DSS
PHD 2
Star Tools / Sky Tools

Subject: IC 1805
SHO Heart

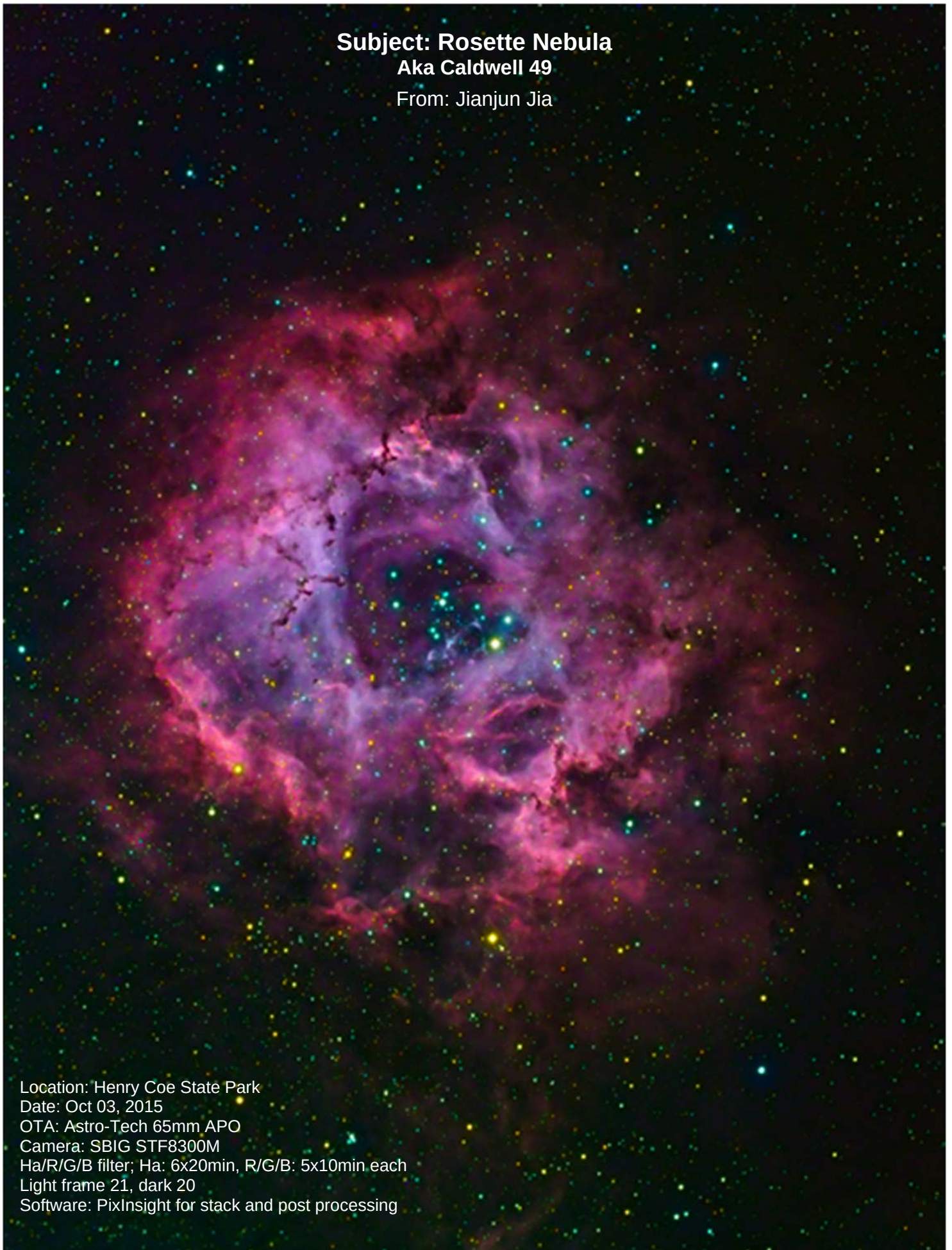
From Rob Pfile

Location: Casa de Rob, Oakland, CA
Dates: Oct. 22, 2015, Oct. 30, 2015, Nov. 4, 2015
Telescope: AT6RC
Camera: SBIG STT 8300M
Mount: Astro-Physics Mach1 GTO
Focal reducer: TeleVue TRF-2008 0.8x
Software: PixInsight, AstroTortilla, PHD guiding,
Astrometry.net
Filters: OIII 5nm, SII 5nm, Ha 5nm
Resolution: 3273x2460
Duration: 29.3 hours

Subject: Rosette Nebula

Aka Caldwell 49

From: Jianjun Jia



Location: Henry Coe State Park

Date: Oct 03, 2015

OTA: Astro-Tech 65mm APO

Camera: SBIG STF8300M

Ha/R/G/B filter; Ha: 6x20min, R/G/B: 5x10min each

Light frame 21, dark 20

Software: PixInsight for stack and post processing

Subject: M20, Trifid Nebula

From: Mark Striebeck

Date: Sept. 26, 2015

Locations: GSSP; Adin, California & Orange Spring Ranch, Mariposa, CA

Imaging telescope: Takahashi TOA-130

Imaging camera: FLI MLx694

Mount: Astro-Physics Mach 1 GTO, Bisque MyT MyT

Guiding telescope: Borg Mini 50mm Guidescope

Guiding camera: Starlight Xpress Lodestar

Focal reducer: TOA-645 Super Reducer

Software: Pixinsight 1.8; Main Sequence Software Sequence Generator Pro

Resolution: 1241x912

Integration: 5.4 hours



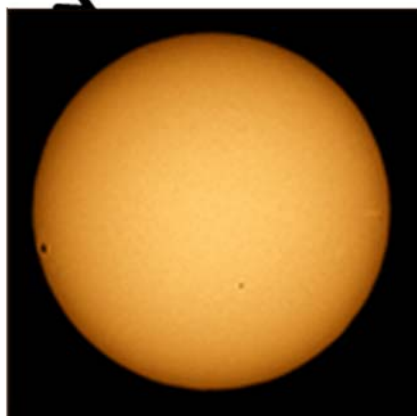


Kid Spot Jokes:

- ♦ **When is the moon the heaviest?**
When it's full.
- ♦ **Why did the sun go to school?**
To get brighter.

Kid Spot Mercury Quiz:

1. How close is Mercury to the Sun?
2. Mercury has active volcanoes. True or False?
3. What is the name of the first spacecraft to orbit Mercury?
4. How many days does it take for Mercury to go around the Sun?
5. How long does it take for Mercury to complete a single rotation?



2006 Transit of Mercury
Photo Credit: Wikipedia

Did You Know?

Source: Astronomy Magazine March 2015 issue "500 Coolest Things"

Early people used Stonehenge as an observatory starting around 2300 B.C.

To escape Earth's gravity, a spacecraft must travel more than 25,008 mph, or near Mach 33.

Three quarters of all normal matter in the universe is hydrogen.

In 1967, John Dobson created a simple and inexpensive alt-azimuth mounting for Newtonian reflector telescopes that later observers call the Dobsonian mount.

From Earth, the Sun appears 30 times larger and 900 times brighter than Neptune.

Astronomers have used the Hubble Space Telescope to estimate that a whopping 125 billion galaxies populate the visible universe.

Some 4.7 million years ago the currently magnitude 1.5 star Adhara (Epsilon Canis Majoris) was more than 350 light-years closer to our solar system and 160 times brighter than it is now.

The 60 ton Hoba iron meteorite in Namibia is the world's largest intact meteorite.

German scientist Johann Wilhelm Ritter discovered ultra-violet radiation in 1801.

Kid Spot Quiz Answers:

1. 26 million miles (0.39 AU)
2. False
3. Messenger
4. 88 days
5. 59 days

From the Board of Directors

Announcements

Officer Elections

At the March 19, 2016 meeting the board elected unanimously club officers for the next year as follows:

President: Dave Ittner
Vice President: Ed Wong
Treasurer: Rob Jaworski
Secretary: Rob Chapman

Binocular viewing first session for 2016 is scheduled for May 28th at OSA's Coyote Valley site.

The Calstar 2016 event is scheduled for 9/29 – 10/2, again at Whisper Canyon.

July 16, 2016 is Movie Night!

SJAA's Yosemite Public Star Party will be held on August 12th & 13th. Contact Jim Van Nuland for more details.

The SJSS Deep Sky Scavenger Hunt (DSSH) is scheduled for June 4th at Pinnacles West. The event has limited places available; contact Ed Wong regarding reservations.

Board Meeting Excerpts

February 20, 2016

In attendance

Dave Ittner, Lee Hoglan, Bill O'Neil, Glenn Newell, Rob Jaworski, Vini Carter, Ed Wong, Teruo Utsumi
Guests: Bruce Braunstein, Rob Chapman, Tom Piller, Gary Chock

City of San Jose Survey

Rob drafted an announcement to publicize the survey from the city of San Jose regarding use of Houge Park. Rob will send out the announcement.

Volunteers

Teruo put together descriptions of roles for some operational aspects of the club and put them in a binder. Teruo requested everyone look through the binder and provide feedback, as well as add other roles.

Coders

Teruo reported on the Coders effort. We have two possibilities for membership management: 1) SMF + Paypal

plugin or 2) Tendenci, an open source software package.

Loaner Program

Muditha Kanchana (Kanch) has taken over the Loaner Program from Manoj Koushik.

Imaging Loaner Setup

Glenn presented his proposal for an imaging setup for the Loaner Program. Concern about the cost (several \$K) was raised. Glenn to re-evaluate proposal. Discussion to be continued at a later date.

Imaging SIG

Bruce Braunstein has taken over the Imaging SIG from Mark Striebeck.

March 19, 2016

In attendance

Dave Ittner, Ed Wong, Lee Hoglan, Bill O'Neil, Glenn Newell, Vini Carter, Teruo Utsumi

Absent: (excused) Rob Jaworski (proxy to Dave)

Guests: Werner Kuballa, Rob Chapman

Open Board Seat Filled

Bill O'Neil nominated Rob Chapman to fill the open board seat. The nomination was approved by all.

Imaging Loaner Equipment

Glenn modified his proposal to make use of existing equipment in the loaner program and to have the equipment for demo/hands-on, not for loan. The motion to accept Glenn's proposal / budget for the equipment was approved by all.

Summer Lick Tour

Bill O'Neil proposed SJAA sponsor another night viewing trip to Lick Observatory, similar to last year's trip. One request we've had is to hold the trip on a weekend night, but it's easier for Lick to accommodate a weeknight and there appears to be sufficient interest to fill the number of slots. Limited to max 30. Bill to contact Lick.

April 24, 2016

In attendance

Rob Chapman, Lee Hoglan, Bill O'Neil, Glenn Newell, Rob Jaworski, Vini Carter, Ed Wong, Teruo Utsumi
Absent (excused): Dave Ittner
Guests: Nick and Amy

Memberships:

Ed Wong is handing off management of memberships to Anand.

Binocular Viewing Program:

Ed purchased two sets of binoculars and tripods. Members have been trying them out and writing up reviews/reports.

Imaging Program Purchase Report:

Glen has completed the purchase of the gear for the two astrophotography rigs. He is very close to the budget. Minor accounting to sort out with Rob J. Rigs setup for board to take a look. Everything looks great!

Lick Observatory Event:

Sold Out. There was discussion and general agreement that the event should be held more often due to its popularity. Further discussion was deferred to the next meeting.

Swap Meet:

Everyone liked the improved format for the March 2016 swap meet and felt that it went well. The Board discussed options on expanding the space available for sellers along with whether a summer swap meet was necessary. The general consensus was against a summer swap meet for various reasons. Dave Ittner to review and provide a recommendation for Board approval.

Binocular Loaner Program:

Ed Wong requested and received authorization to purchase padded cases for tripod and parallelogram mounts for the Binocular equipment.

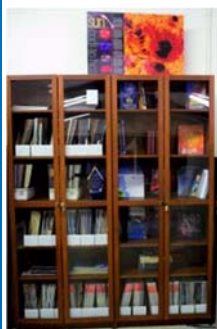
Membership Database Development

Teruo Utsumi reported that the membership database software package (Tendency) is now functional. Teruo has finished beta testing the new package and is now integrating the membership payment system into membership database updates.

SJAA Astronomical Event Calendar

Teruo Utsumi is developing an astronomical calendar scheduling app for the SJAA website to help automate scheduling events onto the sjaa.net calendar as well as sjaa.meetup; signing up volunteers, sending out reminder emails, etc.

SJAA Library!



SJAA offers another wonderful resource; a library with good astronomy books and DVDs available to all of our members that will interest all age groups and especially young children who are budding astronomers!

Please check out our wish list on the SJAA webpage:

<http://www.sjaa.net/sjaa-library/>

Telescope Fix It Session

Fix It Day, sometimes called the Telescope Tune Up or the Telescope Fix It program is a real simple service the SJAA offers to members of the community for free, though it's priceless. The Fix It session provides a place for people to come with their telescope or other astronomy gear problems and have them looked at, such as broken scopes whose owners need advice, or need help with collimating a telescope.

<http://www.sjaa.net/programs/fixit/>

Solar Observing

Solar observing sessions, headed up by Bill O'Neil, are usually held the 1st Sunday of every Month from 2pm - 4pm at Houge Park weather permitting. Please check SJ Astronomy Meetup for schedule details as the event time / location is subject to change:

<http://www.meetup.com/SJ-Astronomy/>

Quick START Program

The Quick START Program, headed up by Dave Ittner, helps to ease folks into amateur astronomy. You have to admit, astronomy can look exciting from the outside, but once you scratch the surface, it can get seemingly complex in a hurry. But it doesn't have to be that way if there's someone to guide you and answer all your seemingly basic questions.

The Quick Start sessions are generally held every other month.

<http://www.sjaa.net/programs/quick-start/>

Intro to the Night Sky

The Intro to the Night Sky session takes place monthly, in conjunction with first quarter moon and In Town Star Parties at Houge Park. This is a regular, monthly session, each with a similar format, with

only the content changing to reflect what's currently in the night sky. After the session, the attendees will go outside for a guided, green laser tour of the sky, along with a club telescope to get a better look at celestial objects.

<http://www.sjaa.net/programs/beginners-astronomy/>

Loaner Program

Muditha Kanchana (Kanch) has taken over the Loaner Program from Manoj Koushik. This program is for SJAA members to evaluate equipment they are considering purchasing or are just curious about. Check out the growing list of equipment below. Please note that certain items have restrictions or special conditions that must be met.

If you are an SJAA member and an experienced observer or have been through the SJAA Quick START program please fill this form to request a particular item. Please also consider donating unused equipment.

<http://www.sjaa.net/programs/loaner-telescope-program/>

Astro Imaging Special Interest Group (SIG)

(SIG), In February, Bruce Braunstein agreed to lead the SIG group. SIG has a mission of bringing together people who have an interest in astronomy imaging, or put more simply, taking pictures of the night sky. The Imaging SIG meets roughly every month at Houge Park to discuss topics about imaging. The SIG is open to people with absolutely no experience but want to learn what it's all about, but experienced imagers are also more than welcome, indeed, encouraged to participate. The best way to get involved is to review the postings on the SJAA Astro Imaging mail list in Google Groups.

<http://www.sjaa.net/programs/imaging-sig/>

Astro Imaging Workshops & Field Clinics

Not to be confused with the SIG group this newly organized program championed by Glenn Newell is a hands on program for club members, who are interested in astro-photography, to have a chance of seeing what it is all about. Workshops are held at Houge Park once per month and field clinics (members only) once per quarter at a dark sky site. Check the schedule and contact Glenn Newell if you are interested.

School Star Party

The San Jose Astronomical Association conducts evening observing sessions (commonly called "star parties") for schools in mid-Santa Clara County, generally from Sunnyvale to Fremont to Morgan Hill.

Contact SJAA's Jim Van Nuland (Program Coordinator) for additional information.



<http://www.sjaa.net/programs/school-star-party/>

SJAA Contacts

President:	Dave Ittner
Vice President:	Ed Wong
Treasurer:	Rob Jaworski
Secretary:	Rob Chapman
Director:	Lee Hoglan
Director:	Bill O'Neil
Director:	Teruo Ustsumi
Director:	Vini Carter
Director:	Glenn Newell
Memberships:	Anand Rajagopalan
Intro to the Night Sky:	David Grover
Fix-it Program:	Vini Carter
Imaging SIG:	Bruce Braunstein
Hands On Imaging:	Glenn Newell
Library:	Sukhada Palav
Loaner Program:	Muditha Kanchana
Ephemeris Newsletter:	
Editor:	Sandy Mohan
Prod. Editor:	Tom Piller
Questions:	Lee Hoglan
Quick START	Dave Ittner
Solar:	Bill O'Neil
School Events:	Jim Van Nuland
Speakers:	Sukhada Palav
E-mails:	http://www.sjaa.net/contact

SJAA Ephemeris, the newsletter of the San Jose Astronomical Association, is published quarterly.

Articles for publication should be submitted by not later than the 20th of the month of February, May, August and November. (earlier is better).

San Jose Astronomical Association
P.O. Box 28243
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San Jose Astronomical Association Annual Membership Form

P.O. Box 28243 San Jose, CA 95159-8243

Membership Type:

- ☐ **New** ☐ **Renewal** (Name only if no corrections)
- ☐ **\$20** Regular Membership with **online** Ephemeris
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The newsletter is always available online at: <http://www.sjaa.net/sjaa-newsletter-ephemeris/>

Questions? Send e-mail to: sjaamemberships@gmail.com

Bring this form to any SJAA Meeting or send to the address (above). Make checks payable to "SJAA", or join/renew at: <http://www.sjaa.net/join-the-sjaa/>

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